B1 — 24 V DC / 100 W MAX HEATER DESIGN TARGET

J2
24V IN / + PGND

F1
6.3A / 125VDC / 045106.3MRL

Q2
DMP6023LE-13 / reverse input

D3
BZT52C12 / 12V gate clamp

D2
SMBJ24CA / transients NOT qualified

J3
HEATER / + switched return

R8
100k

D4
STPS5L60U / cable flyback

C8
100uF / 50V / EEUFR1H101

#FLG01
PWR_FLAG

#FLG02
PWR_FLAG

#FLG03
PWR_FLAG

#FLG04
PWR_FLAG

#FLG05
PWR_FLAG

Regulated 24 V supply only. F1 is a short-circuit protection candidate, not a 100 W limiter.
PMOS: drain = fused input, source = protected output. Zener cathode at source.
TVS clamping/transient margin of the purchased 32 V module is NOT qualified; no industrial surge claim.

Integrated LM2596 fixed 5V. NT1 joins PGND/GND at input star; heater return stays separate.
F2 remains 1A. USB source sharing and thermal/fault qualification still pending.

F2
1A / 125VDC / 0451001.MRL

U2
AP2112K-3.3

C1
1uF / 16V X7R

C2
1uF / 10V X7R

Q1
IRLZ44NPBF / gate 5 V

R3
22R

J1
A-B POWER / XH-2

J6
A-B I2C / XH-3

J7
A-B CTRL / XH-6

B3 — PWM AND ENABLE / 5 V GATE DRIVE

U3
SN74LVC1G08DBVR

U5
UCC27517DBVR

R1
47k

R2
47k

C3
100nF / 10V X7R

C6
100nF / 16V X7R

C7
1uF / 16V X7R

R4
100k

Integrated LM2596 fixed 5V. NT1 joins PGND/GND at input star; heater return stays separate.
F2 remains 1A. USB source sharing and thermal/fault qualification still pending.

B4 — NTC / ADS1115 ADDRESS 0x49

J4
TEMP / NTC 15k B3920

R5
8.45k 1%

R6
1k

C4
100nF / 10V X7R

U4
ADS1115IDGSR / 0x49

C5
100nF / 10V X7R

R7
10k

24 V / 100 W at 500 Hz PWM is a PROTOTYPE target. Independent thermal cutoff still required for unattended use.
Firmware must handle open/short NTC, stale ADC, fault shutdown and explicit enable. No firmware changes in this revision.

DIAGNOSTICS — RAIL PRESENCE AND HEATER GATE DRIVE

D5 / 24V

R9
15k / 1% / 0.125W

D5
APT2012LZGCK

D6 / 5V

R10
1.5k / 1% / 0.1W

D6
APT2012LZGCK

D7 / 3V3

R11
470R / 1% / 0.1W

D7
APT2012LZGCK

D8 / HEAT

R12
2.2k / 1% / 0.1W

D8
APT2012LSECK/J4-PRV

Green LEDs show rail presence only, not regulated voltage accuracy or ESP32 execution.
HEAT (B/D8): 5 V driver output after PWM AND ENABLE; brightness follows duty, NOT measured heater current.
NTC on B/J4: 1 = +3V3_B, 2 = NTC_RAW; passive two-wire 15k B3920 probe, either wire orientation.

B5 — INTEGRATED BUCK / FIXED 5V / SYSTEM BUDGET 0.8A

U1
LM2596S-5.0/NOPB

L1
100uH / SRP1770TA-101M

D9
SS36 / 60V 3A

C9
100nF / 50V X7R

C10
330uF / 35V

R13
56m / 1% / 0.25W

NT1
GND STAR / NO COMPONENT

C8 is the local 100uF/50V input reservoir; C9 is HF bypass. C10 negative returns through R13.
LM2596 tab/pin 3 = PGND. Pin 5 hard-wired low (buck always enabled with 24V present).
NT1 is a copper net tie, not an isolation barrier. No external XL4015 or J8/J9.
Matched electrical study: hardware/power_study/lm2596. Physical startup, fault and thermal tests remain required.

BENCH TEST PADS — BACK SIDE B.Cu — COPPER ONLY, DO NOT POPULATE

TP1 PGND
TP2 VIN_PROTECTED
TP3 +5V_BUCK
TP4 +5V_AB
TP5 +3V3_B
TP6 NTC_RAW
TP7 NTC_ADC
TP8 PWM0
TP9 ENABLE
TP10 FAULT_N
TP11 GATE_DRV
TP12 GND

PROTOTYPE — ENGINEERING PREPARED — ASSEMBLER APPROVAL REQUIRED

Laboratory hardware platform

Sheet: /
File: pcb_b_temperature.kicad_sch

Title: PCB B — Temperature / 24 V 100 W

Size: User | Date: 2026-11-05 | **Rev: 0.5.1**
KiCad E.D.A. 10.0.6 | Id: 1/1